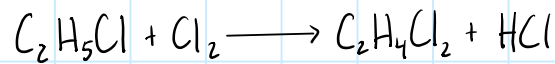
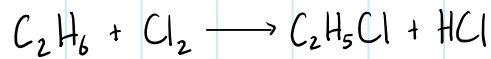
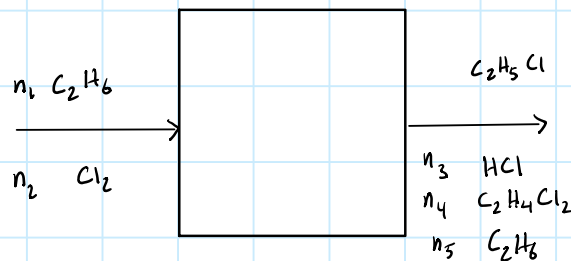


①



- a) To maximize selectivity of the monochloroethane production relative to the dichloroethane production, the reactor should be designed for a low conversion of ethane because that lowers the amount of monochloroethane so that the undesired side reaction won't occur. To make the process more economical, the undesired side reaction should not occur. ✓

b)



$$\text{DoF} = (4a + 4b) - (5a + 5b + 5c + 5d)$$

$$\text{DoF} = (6 + 0) - (0 + 0 + 0 + 3) = 3 \quad \times \quad -3 \text{ pts}$$

5

2

c)

Basis: 100 mol $\text{C}_2\text{H}_5\text{Cl}$ produced

$$100 \text{ mol } \text{C}_2\text{H}_5\text{Cl} = 14 n_4$$

$$n_4 = \frac{100}{14} = 7.14 \text{ mol } \text{C}_2\text{H}_4\text{Cl}_2$$

$$n_4 = \frac{100}{14} = 7.14 \text{ mol } C_2H_4Cl_2$$

15% conversion

$$(1 - 0.15)n_1 = n_5$$

$$0.85 n_1 = n_5$$

$$n_1 = \frac{n_5}{0.85}$$

$$C: 2n_1 = 2(100 \text{ mol}) + 2n_4 + 2n_5$$

$$2\left(\frac{n_5}{0.85}\right) = 200 + 2(7.14) + 2n_5$$

$$2.35 n_5 = 214.29 + 2n_5$$

$$0.35 n_5 = 214.29$$

$$n_5 = 607.14 \text{ mol } C_2H_6$$

$$n_1 = \frac{607.14}{0.85} = 714.29 \text{ mol } C_2H_6$$

$$H: 6n_1 = 5(100) + n_3 + 4n_4 + 6n_5$$

$$6(714.29) = 500 + n_3 + 4(7.14) + 6(607.14)$$

$$4285.71 = 4171.43 + n_3$$

$$114.29 \text{ mol} = n_3$$

HCl

$$Cl: 2n_2 = 100 + n_3 + 2n_4$$

$$2n_2 = 100 + 114.29 + 2(7.14)$$

$$n_2 = 114.29 \text{ mol } Cl_2$$

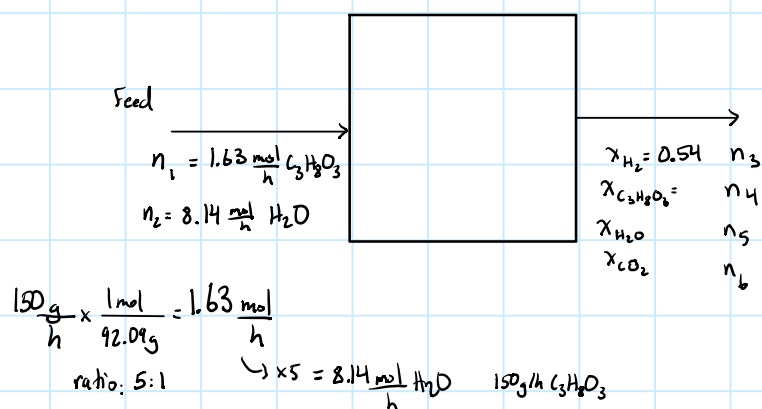
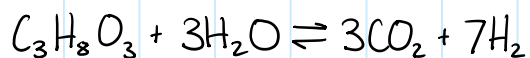
$$\text{Feed Ratio} = \frac{\text{mol } Cl_2}{\text{mol } C_2H_6} = \frac{114.29}{714.29} = 0.16$$

$$\text{Fractional Yield} = \frac{n_{C_2H_5Cl}}{n_{max}} = \frac{100 \text{ mol}}{114.29} = 0.875 \rightarrow 87.5\%$$

d) Since there's no Cl_2 in the product, that means all of the Cl_2

d) Since there's no Cl_2 in the product, that means all of the Cl_2 reacted but not only with ethane. It reacted further with the products creating heavier compounds.

(2)



Let x = fractional conversion of glycerine Use basis 1 mol $\text{C}_3\text{H}_8\text{O}_3$

Let X = fractional conversion of glycerine Use basis 1 mol $C_3H_8O_3$

	Initial (mol)	Change (mol)	Exit (mol)
$C_3H_8O_3$	1	$-X$	$1-X$
H_2O	5	$-3X$	$5-3X$
CO_2	0	$+3X$	$3X$
H_2	0	$+7X$	$7X$

$$n_{out} = (1-X) + (5-3X) + 3X + 7X$$

$$n_{out} = 6 + 6X = 6(1+X)$$

$$H_2 \text{ mole fraction: } y_{H_2} = \frac{7X}{6+6X}$$

$$\frac{7X}{6+6X} = 0.54$$

$$7X = 0.54(6+6X)$$

$$7X - 3.24X = 3.24$$

$$3.76X = 3.24$$

$$X = 0.8617 \rightarrow 86.17\%$$

$$K_a = 55 \text{ atm}^6$$

$$K_a = \frac{(P_C)^3 (P_H)^7}{P_G (P_W)^3}$$

$$P_i = x_i P$$

reactor operates at 1.2 atm

$$C_3H_8O_3: 1-X = 1-0.8617 = 0.13$$

$$H_2O: 5-3X = 5-3(0.8617) = 2.41$$

$$CO_2: 3X = 3(0.8617) = 2.59$$

$$H_2: 7X = 7(0.8617) = 6.03$$

$$n_{tot} = 11.17 \text{ moles}$$

mole fractions

$$X_{H_2} = \frac{6.03}{11.17} = 0.539 \approx 0.54 \quad P_{H_2} = 0.54 \times 1.2 \text{ atm} = 0.648$$

$$X_{CO_2} = \frac{2.59}{11.17} = 0.23 \quad P_{CO_2} = 0.23 \times 1.2 \text{ atm} = 0.2777$$

$$X_{H_2O} = \frac{2.41}{11.17} = 0.216 \quad P_{H_2O} = 0.216 \times 1.2 \text{ atm} = 0.259$$

$$X_{C_3H_8O_3} = \frac{0.13}{11.17} = 0.012 \quad P_{C_3H_8O_3} = 0.012 \times 1.2 \text{ atm} = 0.0144$$

$$K_a = \frac{(0.2777)^3 (0.648)^7}{0.0144 (0.259)^3} = 3.96 \ll 55 \quad \text{Not at equilibrium} \quad \checkmark$$

To do better, you can decrease the pressure so the reaction can shift towards the higher volume (product side) or remove H_2 that's produced. ✓

If the reactor outlet was at equilibrium,

$$\begin{array}{ll} C_3H_8O_3: & 1.63 - X \\ H_2O: & 8.14 - 3X \\ CO: & 3X \\ H_2: & 7X \end{array} \quad \left. \begin{array}{l} \text{Total} = (1.63 - X) + (8.14 - 3X) + 3X + 7X \\ \text{Total} = 9.77 + 6X \end{array} \right\}$$

$$K_a = \frac{\left(\frac{3X}{9.77+6X}\right)^3 (1.2)^3 \left(\frac{7X}{9.77+6X}\right) (1.2)^7}{\left(\frac{1.63-X}{9.77+6X}\right) (1.2) \left(\frac{8.14-3X}{9.77+6X}\right)^3 (1.2)^3} = 55$$

25/25

$$K_a = \frac{6.64 \times 10^7 X^{10}}{(1.63 - X)(8.15 - 3X)^3(9.77 + 6X)^6} = 55$$

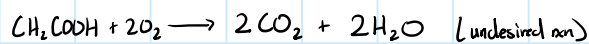
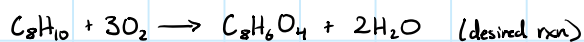
Through trial error, $X = 1.575$

$$\text{mol of } H_2 = \frac{7X}{9.77 + 6X} = \frac{7(1.575)}{9.77 + 6(1.575)} = 0.573$$

$$\text{conversion of } C_3H_8O_3 = \frac{1.575}{1.63} = 0.9662 \rightarrow 96.62\%$$

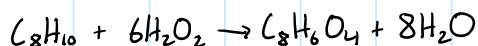
③ 4.49

Conventional process



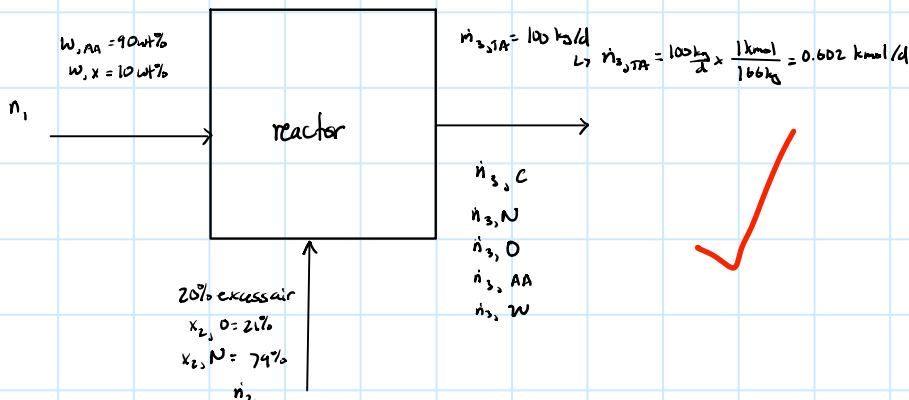
$$\text{atom economy} = \frac{M_{C_8H_6O_4}}{3M_{O_2} + M_{C_8H_{10}}} = \frac{166}{3(32) + 106} = 0.822 \leftarrow \text{has better atom economy}$$

New process



$$\text{atom economy} = \frac{M_{C_8H_6O_4}}{6M_{H_2O_2} + M_{C_8H_{10}}} = \frac{166}{6(34) + 106} = 0.535$$

AA = acetic acid X = Xylene W = H₂O TA = terephthalic acid O = oxygen N = nitrogen C = CO₂



$$Dof = (4a + 4b) - (5a + 5b + 5c + 5d) \\ = (10 + 2) - (1 + 2 + 2 + 7) = 0$$

species mole balance:

$$TA: \dot{n}_{3,TA} = \dot{n}_{1,TA} + \dot{S}_1 \rightarrow \dot{S}_1 = \dot{n}_{3,TA} = 0.602 \frac{\text{kmol}}{\text{d}}$$

$$XY: \dot{n}_{3,XY} = \dot{n}_{1,XY} - \dot{S}_1 \rightarrow \dot{n}_{1,XY} = \dot{S}_1 = 0.602 \frac{\text{kmol}}{\text{d}}$$

$$\frac{\dot{n}_{1,XY}(M_X)}{\dot{m}_1} = 0.1 \rightarrow \frac{0.602 \frac{\text{kmol}}{\text{d}} (106 \frac{\text{kg}}{\text{kmol}})}{\dot{m}_1} = 0.1 \\ \dot{m}_1 = 638.12 \text{ kg/d}$$

$$\dot{m}_{1,AA} = 0.9 \dot{m}_1 = 0.9 (638.12 \text{ kg/d}) = 574.308 \text{ kg/d}$$

$$\dot{n}_{3,AA} = 574.308 \frac{\text{kg}}{\text{d}} \times \frac{1 \text{ kmol}}{60 \text{ kg}} = 9.57 \frac{\text{kmol}}{\text{d}} AA$$

$$\dot{n}_1 = \dot{n}_{1,XY} + \dot{n}_{1,AA} = (0.602 + 9.57) \frac{\text{kmol}}{\text{d}} = 10.172 \text{ kmol/d}$$

$$10\% \text{ is oxidized} \rightarrow \dot{S}_2 = 0.1 \quad \dot{n}_{2,AA} = 0.1 (9.5718 \frac{\text{kmol}}{\text{d}} AA) = 0.957$$

$$\text{From undesired rxn: } \dot{n}_{3,AA} = \dot{n}_{2,AA} - \dot{S}_2 = (9.57 - 0.957) = 8.613 \frac{\text{kmol}}{\text{d}} AA$$

$$\dot{n}_{3,C} = 2 \dot{S}_2 = 2 (0.957 \frac{\text{kmol}}{\text{d}}) = 1.916 \frac{\text{kmol}}{\text{d}}$$

$$\dot{n}_{3,W} = 2 \dot{S}_1 + 2 \dot{S}_2 = 2 (0.602) + 2 (0.957) = 3.118 \frac{\text{kmol}}{\text{d}}$$

$$\dot{n}_{0,10\%} = 3 \dot{S}_1 + 2 \dot{S}_2 = 3 (0.602) + 2 (0.957) = 3.72 \frac{\text{kmol}}{\text{d}}$$

$$\dot{n}_{2,O} = (1 + 0.2) (3.72 \frac{\text{kmol}}{\text{d}}) = 4.464 \frac{\text{kmol}}{\text{d}} \quad \frac{\dot{n}_{2,O}}{\dot{n}_2} = 0.21$$

$$\dot{n}_{3,O} = \dot{n}_{2,O} - \dot{n}_{0,10\%} = 4.464 - 3.72 = 0.766 \frac{\text{kmol}}{\text{d}} \quad \dot{n}_2 = \frac{\dot{n}_{2,O}}{0.21} = \frac{4.464}{0.21} = 21.257 \frac{\text{kmol}}{\text{d}}$$

$$\dot{n}_{2,N} = (0.79) (21.257) = 16.79 \frac{\text{kmol}}{\text{d}} = \dot{n}_{3,N}$$

$$\dot{n}_3 = \dot{n}_{3,TA} + \dot{n}_{3,C} + \dot{n}_{3,W} + \dot{n}_{3,O} + \dot{n}_{3,N} + \dot{n}_{3,AA}$$

$$\dot{n}_3 = 0.602 + 1.916 + 3.118 + 0.766 + 16.79 + 8.613$$

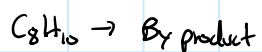
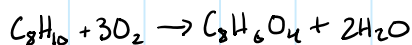
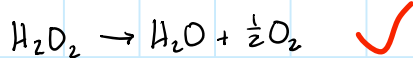
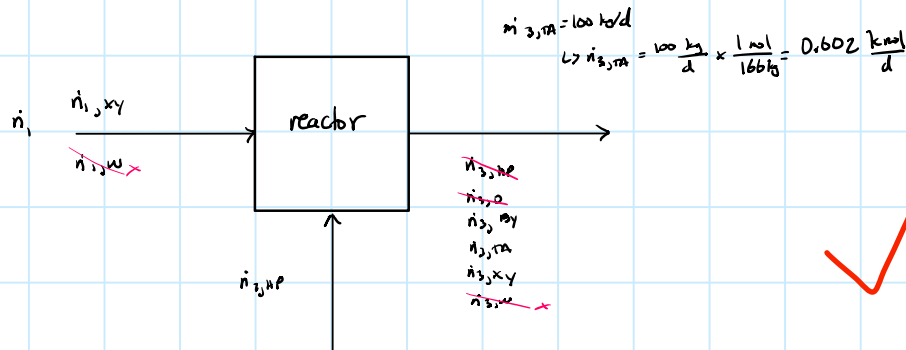
$$\dot{n}_3 = 31.811 \frac{\text{kmol}}{\text{d}}$$

Fractional conversion: mol xylene convert to TA - $\dot{n}_{2,TA}$ none

$$\dot{n}_3 = 31.811 \frac{\text{kmol}}{\text{d}}$$

$$\text{Fractional Conversion: } \frac{\text{mol xylene convert to TA}}{\text{mol xylene feed}} = \frac{\dot{n}_{3,TA}}{\dot{n}_{1,XY}} = \frac{0.602}{0.602} = 1$$

New process



$$\text{DoF} = (9+3) - (1+0+2+6) = 3 \text{ under specified}$$

Assume O_2 & H_2O_2 react completely 100% & discard the amount of water

$$\text{DoF} = (6+3) - (1+0+2+5) = 0$$

$$\text{selectivity} = \frac{\dot{n}_{3,TA}}{\dot{n}_{3,TA} + \dot{n}_{3,X}} = 0.9 \quad \dot{n}_{3,TA} = \frac{0.602}{0.9} = 0.669 \frac{\text{kmol}}{\text{d}}$$

$$\text{fractional yield} = \frac{\dot{n}_{3,TA}}{\dot{n}_{3,X}} = 0.7 \rightarrow \dot{n}_{3,X} = \frac{0.602}{0.7} = 0.86 \frac{\text{kmol}}{\text{d}}$$

$$\text{TA: } \dot{n}_{3,TA} = 0 + \dot{S}_2 \rightarrow \dot{S}_2 = 0.602 \frac{\text{kmol}}{\text{d}}$$

$$\text{X: } \dot{n}_{3,XY} = \dot{n}_{1,XY} - \dot{n}_{3,TA} = 0.86 - 0.669 = 0.191 \frac{\text{kmol}}{\text{d}}$$

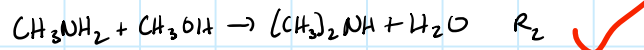
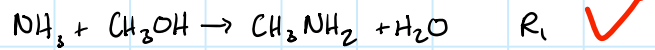
$$\text{O: } \dot{n}_{3,O} = \frac{1}{2}\dot{S}_1 - 3\dot{S}_2 \rightarrow 0 = \frac{1}{2}\dot{S}_1 - 3\dot{S}_2 \rightarrow \dot{S}_1 = 3.612 \frac{\text{kmol}}{\text{d}}$$

$$\text{HP: } \dot{n}_{3,HP} = \dot{S}_1 = 3.612 \frac{\text{kmol}}{\text{d}}$$

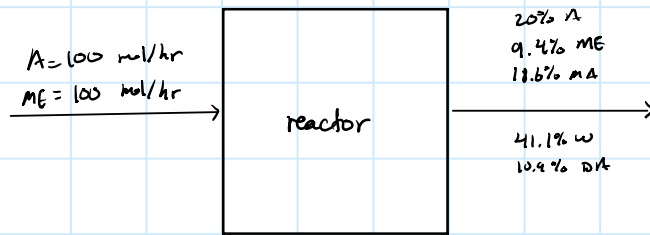
I wouldn't suggest Connie's process because the fractional yield of the conventional process is greater.

④ 4.50

④ 4.50



$A = \text{NH}_3$ $ME = \text{CH}_3\text{OH}$ $MA = \text{CH}_3\text{NH}_2$ $W = \text{H}_2\text{O}$ $DA = (\text{CH}_3)_2\text{NH}$



Total mol balance @ S.S.

$$n_{\text{out}} = n_{\text{in}} + \dot{S}_1(1+1-1-1) + \dot{S}_2(1+1-1-1)$$

$$n_{\text{out}} = n_{\text{in}} = 100 \frac{\text{mol}}{\text{hr}} \text{ ME} + 100 \frac{\text{mol}}{\text{hr}} \text{ A} = 200 \frac{\text{mol}}{\text{hr}}$$

a) $f_{CM} = \frac{n_{\text{min}} - n_{\text{out}}}{n_{\text{min}}} = \frac{100 - 0.094(200)}{100} = 0.81$ fractional conversion of methanol $\checkmark$

b) yield of MA based on M = $\frac{0.186(200)}{100} = 0.37$ $\checkmark$

c) selectivity of MA based on M = $\frac{0.186(200)}{100 - (0.094 \times 200)} = 0.46$ $\checkmark$

d) $n_{A,\text{out}} = n_{A,\text{in}} + n_{A_2} \rightarrow 0.2(200) = 100 - \dot{S}_1$
 $\dot{S}_1 = 60 \text{ mol/hr}$ $\checkmark$

e) $n_{DA,\text{out}} = n_{DA,\text{in}} + \dot{S}_2$ $\dot{S}_2 = 0.109(200) = 21.8$

$k_1 = 4$ $k_1 = \frac{y_{MA} y_W}{y_{ME} y_A} = \frac{0.186(200) \times 0.411(200)}{0.2(200) \times 0.094(200)} = 4.066 \approx 4$ $\checkmark$

$k_2 = \frac{y_{DA} y_W}{y_{MA} y_{ME}} = \frac{0.109(200) \times (0.411 \times 200)}{0.186(200) \times 0.094(200)} = 2.56 \approx 2.5$

$$K_2 = \frac{Y_{DA} Y_W}{Y_{MA} Y_{ME}} = \frac{0.109(200) \times (0.411 \times 200)}{0.186(200) \times (0.094 \times 200)} = 2.56 \approx 2.5$$

The reactor is at equilibrium

d) $\dot{n}_{ME,out} = \dot{n}_{ME,in} - \dot{S}_1 - \dot{S}_2$

$$\dot{n}_{MA,out} = \dot{S}_1 - \dot{S}_2$$

$$\dot{n}_{W,out} = \dot{S}_1 + \dot{S}_2$$

$$\dot{n}_{DA,out} = \dot{S}_2$$

$\dot{n}_{A,in}$	$\dot{n}_{ME,in}$	$\dot{S}_1$	$\dot{S}_2$	$\lambda_{MA,out}$	MA/DA	Yield MA from A	Yield MA on ME
100	50	36.8	8.1	12	3.5	0.22	0.57
100	100	59	21.8	18.8	1.7	0.376	0.576
100	200	83.2	47.4	11.9	0.75	0.36	0.18

at $\dot{n}_{A,in} = 100$ & $\dot{n}_{ME,in} = 100$,

$$DA \rightarrow 0.109(200) = \dot{S}_2 \quad \dot{S}_2 = 21.8$$

$$MA \rightarrow 0.186(200) = \dot{S}_1 - 21.8 \quad \dot{S}_1 = 59$$

$$W \rightarrow 59 + 21.8 = 80.8$$

$$ME \rightarrow 100 - 59 - 21.8 = 19.2$$